

A Lightweight and Robust Two-Stage Acoustic Pipeline for Embedded Artillery Launch Detection

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Abstract—Acoustic artillery sensing in the field must tolerate background noise while running on low-power embedded hardware. We study the post-trigger classification stage of a lightweight two-stage pipeline for artillery launch detection. The main corpus contains 706 clean labeled events: 62 artillery launches and 644 non-launch events. We train the classifier with an expanded non-launch corpus, class-imbalance mitigation, and train-time augmentation, and we evaluate perturbations applied in the mel-frequency cepstral coefficient (MFCC) domain. We report these perturbations as nominal MFCC-domain signal-to-noise ratio (SNR) levels, not as waveform-domain acoustic SNR. Deployment on an ESP32-S3 with TensorFlow Lite Micro preserves high classification performance under the tested conditions, with Matthews correlation coefficient (MCC) values of 0.9912 at 30 dB and 0.8009 at 5 dB. These results support the use of a lightweight embedded post-trigger classifier in a two-stage detection pipeline, although the robustness effect of quantization is empirical rather than a proven causal mechanism.

Index Terms—acoustic detection, artillery launch, MFCC, CNN, TensorFlow Lite, embedded AI, robustness, feature-space perturbation

I. INTRODUCTION

Passive acoustic sensing can support early warning in contested environments where active systems are expensive, power-demanding, or easier to locate. Recent high-intensity conflicts have increased the need for low-cost passive systems that can detect artillery launches without emitting signals of their own. A practical acoustic system for this task must satisfy two constraints: (i) robustness to background noise and event variability and (ii) feasible execution on low-power embedded hardware.

Existing acoustic weapon-detection pipelines usually separate continuous-event triggering from the interpretation of short post-trigger signal windows. Triggering often relies on thresholding, energy-envelope analysis, time-of-arrival logic, or multi-sensor consistency checks. Post-trigger classification then uses engineered time-frequency or cepstral descriptors and, more recently, CNN-style classifiers [1]–[4]. For embedded artillery-launch sensing, the key question is whether a compact classifier remains stable under controlled MFCC-domain perturbation and after quantized deployment on low-power hardware.

We adopt a two-stage processing architecture. A streaming trigger first identifies candidate impulsive events, and a post-

trigger classifier then processes short acoustic segments. We treat the trigger, data acquisition, and circular-buffer logic as previously validated components and focus on the artillery launch classifier. Within this scope, we test whether a lightweight classifier preserves useful discrimination under controlled MFCC-domain perturbations reported as nominal signal-to-noise ratio (SNR) levels, while still fitting deployment on ESP32-S3.

Our training strategy combines an expanded non-launch corpus, class-imbalance mitigation, and train-time augmentation. We evaluate whether this design maintains near-clean behavior as perturbation increases. We also assess whether the same classifier remains useful after quantization and deployment through TensorFlow Lite Micro on ESP32-S3, using PC-side 8-bit integer (int8) inference as the immediate reference point.

We frame artillery-launch sensing as a two-stage problem and focus the experiments on the post-trigger classifier. We evaluate the classifier on an expanded 706-event corpus under controlled MFCC-domain perturbation with nominal SNR reporting, and we report ESP32-S3 deployment results together with PC-side int8 verification.

II. STATE OF THE ART

A. Classical Acoustic Detection of Impulsive Events

Weapon-related acoustic surveillance has traditionally used thresholding, energy-envelope analysis, time-of-arrival logic, and multi-sensor consistency checks to detect impulsive transients before semantic classification. Operational pipelines usually split artillery and gunshot processing into two tasks: (i) event triggering in continuous audio streams and (ii) post-trigger interpretation of short signal segments. Battlefield-oriented studies show that reliable operation depends on transient triggering that tolerates propagation distortion, changing stand-off distances, and heterogeneous sensor layouts [1], [2], [5], [6]. Trigger quality therefore limits downstream classifier performance.

B. Acoustic Signature of Artillery and Gunfire

Artillery and gunfire recordings do not follow a single invariant waveform. Source type, firing geometry, propagation, and local reflections shape each event. Depending on scenario and range, a recording may include different mixtures of

muzzle blast, shock waves from supersonic projectiles, and delayed arrivals. Fig. 2 shows an example artillery-launch waveform before extraction of the 60 ms post-trigger analysis window. Earlier work captured these effects indirectly through time-frequency descriptors. Long-range studies modeled wave arrivals explicitly and documented sensitivity to atmosphere, terrain, and sensor geometry [1], [5], [6]. These factors create domain shift: models tuned on narrow or clean recording conditions often degrade when transferred to field recordings with stronger multipath, clutter, and nonstationary background noise.

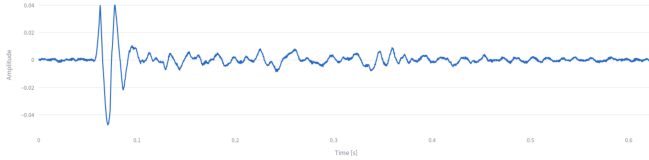


Fig. 1. Example artillery-launch waveform before extraction of the 60 ms post-trigger analysis window.

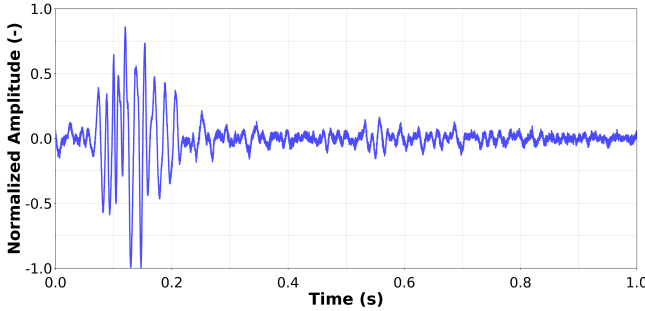


Fig. 2. Example artillery-launch waveform before extraction of the 60 ms post-trigger analysis window.

C. From Detection to Post-Detection Classification

After trigger detection, systems usually classify short post-trigger windows. Classical approaches combine engineered representations, such as wavelet, cepstral, or temporal-envelope features, with shallow classifiers. Later methods use convolutional neural network (CNN) models over spectro-temporal inputs [2], [3], [7]–[9]. Cepstral representations remain attractive because they compactly encode the spectral envelope of short acoustic events [10]. Broader environmental-sound evidence also shows that complementary time-frequency descriptors can matter when transient temporal structure is informative [11]. Comparative evidence from related gunshot-classification experiments suggests that performance depends on the interaction between the feature family and the classifier, not on model choice alone. In those comparisons, mel-frequency cepstral coefficient (MFCC)-based representations remained competitive against related cepstral alternatives for both neural and non-neural classifiers [4]. These findings favor a two-stage design with a robust trigger and a compact post-trigger classifier. In a small-data embedded setting, a

compact representation can be preferable to richer inputs when robustness and deployment cost must be balanced.

This motivates compact CNN topologies that preserve local time-frequency structure while remaining deployable on constrained hardware. A small stack of local convolutions, pooling, and a dense classification head follows the established LeNet-like pattern for compact two-dimensional inputs [12]. In audio classification, analogous CNNs operate on time-frequency representations and learn local spectro-temporal patterns rather than hand-coded feature interactions [3], [13]. For microcontroller-oriented deployments, this established small-footprint design logic is often preferable to deeper architectures when memory, latency, and operator support are limiting factors [14]. In this work, we use a task-specific compact MFCC-CNN adapted to the (58, 12) input geometry and validate it as the second stage of the embedded artillery-launch pipeline.

D. Robustness and Domain-Shift Handling

Recent studies report strong performance under uncontrolled recording conditions, but many results remain tied to dataset-specific priors, acquisition pipelines, and label granularity [4], [7], [8]. In practice, model performance can degrade under perturbation, recording-device mismatch, changes in background scenes, or overlapping sound sources [15]. Clean-set accuracy alone is therefore insufficient. Augmentation strategy, perturbation tolerance, and operating-point calibration become central requirements [9]. For embedded artillery-launch sensing, the relevant test is whether class separability survives perturbations applied to the feature representation when the underlying recordings already come from field conditions.

Feature-domain augmentation provides a relevant methodological precedent. SpecAugment, for example, applies warping and masking directly to log-mel feature inputs rather than to raw audio [16]. Similar feature-level Gaussian noise injection has also been used in noise-robust ASR, where artificial Gaussian noise is added to acoustic feature representations during training [17]. Cepstral-domain processing has likewise been used for noise compensation in robust ASR [18]. These precedents support treating MFCC space as a meaningful domain for robustness-oriented processing. At the same time, waveform-level additive noise and MFCC-domain additive perturbations are not equivalent, because additive noise before feature extraction produces nonlinear distortions in cepstral space [19]. This distinction motivates the nominal feature-space interpretation used in this paper.

E. Embedded Deployment and Runtime Constraints

Edge deployment requires detection quality to be evaluated together with memory footprint, latency, and energy use. Tiny machine learning (TinyML) comparisons show that front-end feature extraction can dominate runtime, so the highest-accuracy model is not always the most deployable one [20]. Raspberry Pi and Tensor Processing Unit (TPU)-class studies further show that moderate accuracy loss may be

acceptable when the system meets real-time constraints and resource budgets [21]. MCU-focused TinyML systems also show that memory, latency, energy, and runtime overhead can dominate architecture and deployment choices on commodity microcontrollers [14], [22]. Together with TensorFlow Lite deployment practice [23], [24] and integer or fixed-point quantization methods for efficient on-device inference [25], [26], these findings motivate validation that separates desktop float32 behavior, quantized desktop verification, and embedded execution.

We therefore address a narrower question than full continuous surveillance: whether a lightweight post-trigger classifier for artillery launch sensing remains robust under controlled perturbation and maintains performance after quantized deployment on an embedded target.

III. METHOD

A. Processing Pipeline

We study the post-trigger classification stage of a two-stage acoustic pipeline. In the full architecture, the first stage follows a median-filter impulse-detection approach: a streaming trigger estimates the local acoustic background, identifies candidate impulsive events, and forwards short buffered segments to a classifier [27]. The trigger, acquisition chain, and online buffering logic are outside the scope of this study and are treated as previously validated components. We evaluate only the second stage, the post-trigger CNN artillery launch classifier.

Each candidate event is represented by a 60 ms waveform segment centered on the detected impulse peak. The segment is asymmetric: 40 % precedes the peak to capture onset structure, and 60 % follows it to retain decay and muzzle-blast content. Fig. 4 shows an example of the resulting waveform segment. We then compute MFCC features from these segments for supervised CNN training and inference.

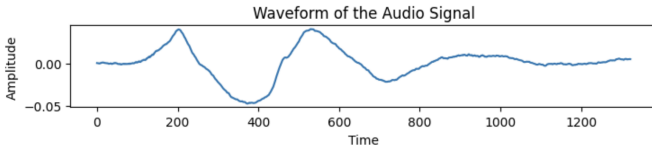


Fig. 3. Example 60 ms peak-centered waveform segment used for MFCC extraction.

B. Field Data Acquisition

The source corpus comes from real artillery measurements and includes both launch and impact acoustic events. We collected the recordings with the field station shown in Fig. 5. The acquisition chain used PreSonus PRM1 microphones connected through a Roland Rubix 44 audio interface, and the signal processing uses 16-bit audio at 22.05 kHz. We model only binary artillery launch classification (artillery launch vs. non-launch) and leave impact-event modeling for future work. The positive class contains only artillery launch events,

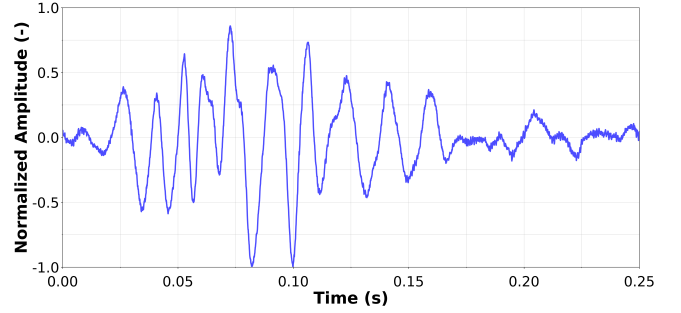


Fig. 4. Example 60 ms peak-centered waveform segment used for MFCC extraction.

whereas the non-launch class contains other impulsive acoustic negatives.

The main clean corpus contains 706 labeled events, summarized in Table I. We expanded the non-launch class beyond the initial development corpus to better reflect operational negatives and to strengthen training under perturbation. An earlier, substantially smaller configuration with approximately 152 labeled events served only as a development-stage precursor and is not the main evaluation corpus reported here.

TABLE I
OVERVIEW OF LABELED ACOUSTIC EVENTS

Subset	Events	Share
Artillery launch	62	8.8 %
Non-launch	644	91.2 %
Main clean corpus	706	100 %



Fig. 5. Field recording station used for artillery dataset acquisition.

C. CNN Topology

The network input is an MFCC tensor of size (58, 12, 1) in `channels_last` format. The classifier first applies a valid Conv2D layer with 32 filters, a 3×3 kernel, stride 1, and rectified linear unit (ReLU) activation, producing a (56, 10, 32) tensor. A 2×2 MaxPooling2D layer with stride 2 reduces

this representation to $(28, 5, 32)$. The second valid Conv2D layer uses 64 filters with the same 3×3 kernel, stride 1, and ReLU activation, producing $(26, 3, 64)$. A second 2×2 MaxPooling2D layer with stride 2 then yields $(13, 1, 64)$. The result is flattened to 832 features, passed through Dense(64) with ReLU, regularized by Dropout(0.5), and classified by a final Dense(1) sigmoid output. The trainable parameter counts are 320 for the first convolution, 18,496 for the second convolution, 53,312 for Dense(64), and 65 for the output layer, for a total of 72,193 trainable parameters; the pooling, flattening, and dropout layers add no trainable parameters. Fig. 6 shows the full architecture. We train the final model for 50 epochs with the Adam optimizer and binary cross-entropy loss. This layout balances representation capacity and deployment cost, following standard convolutional design practice [12].

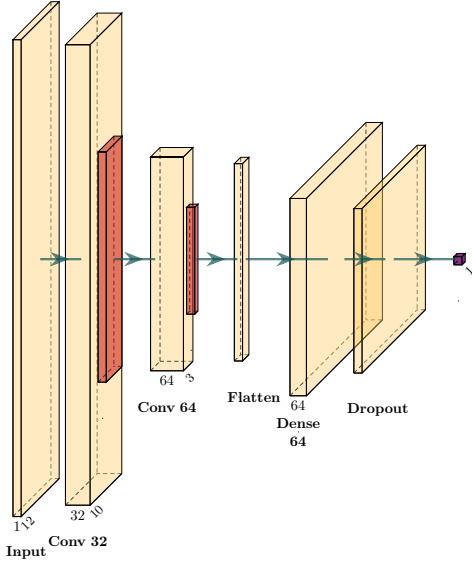


Fig. 6. CNN topology used for post-trigger artillery launch classification of MFCC segments.

D. Robustness Training

We extract MFCCs from 22.05 kHz signals using short-time framing with high temporal resolution. Each 60 ms segment is divided into 2.5 ms frames (approximately 55 samples) with a 1.0 ms hop (approximately 22 samples), corresponding to 1.5 ms overlap (60 %). A Hamming window is applied to each frame to reduce spectral leakage, and the spectrum is evaluated with a 512-point fast Fourier transform ($NFFT = 512$). The mel front end uses 40 mel filters, from which the first 12 MFCC coefficients are retained. This yields the $(58, 12, 1)$ input tensor used by the classifier.

The training strategy combines three elements: (i) an expanded non-launch corpus, (ii) class-imbalance mitigation, and (iii) train-time augmentation of the training split only. This follows the broader use of augmentation for audio classification under limited-data conditions [9]. Gaussian perturbations are applied directly to the precomputed MFCC matrices as

a feature-space input-noise regularization strategy. Since the CNN receives MFCC tensors as its input, this perturbation acts on the representation seen by the network and encourages stable predictions for locally perturbed cepstral representations. Training with input noise is theoretically related to Tikhonov regularization [28], and feature-level Gaussian noise injection has also been used in noise-robust ASR [17]. We report nominal MFCC-domain SNR levels of 30, 20, 10, and 5 dB. In the rest of the paper, we refer to these levels as nominal SNR. These values describe controlled feature-space perturbation and should not be interpreted as waveform-domain acoustic SNR. Additive waveform noise would pass through STFT magnitude computation, mel-filterbank integration, logarithmic compression, and DCT, producing a nonlinear cepstral distortion rather than a simple additive MFCC offset [19].

E. Imbalance and Overfitting Mitigation

Because the corpus is strongly imbalanced, with artillery-launch events representing only about 8.8 % of the 706 labeled samples, training uses class-weighted optimization to reduce majority-class bias, following standard imbalanced-learning practice [29]. The loss assigns a stronger penalty to errors on the minority artillery-launch class, so high nominal accuracy cannot be achieved simply by collapsing predictions toward the majority non-launch class. We treat class weighting as part of the final training strategy, not as an isolated ablation result.

F. Embedded Inference Protocol

After PC training, we export the selected model to TensorFlow Lite Micro [23], [24]. To separate hardware effects from quantization effects, we evaluate three inference regimes: desktop float32 inference, desktop int8 inference with the quantized model, and ESP32-S3 deployment of the same quantized classifier. The embedded implementation uses int8 inference on ESP32-S3 with a statically allocated 80 KB `tensor_arena` for intermediate tensors and a selective `MicroMutableOpResolver` configuration that registers only the operators required by the network. This static-arena and selective-operator design follows the MCU-oriented TensorFlow Lite Micro runtime model. Int8 inference follows standard integer-only practice for reducing memory footprint and enabling efficient execution on resource-constrained processors [24], [25]. Embedded validation uses a host-device protocol: the host computes MFCC tensors, applies the same perturbation settings used in desktop evaluation, transmits the tensors to the embedded target, and collects predicted labels and scores. This host-device transfer is a validation protocol rather than the intended operational pipeline. We use ESP32-S3 as a representative low-power Edge AI platform for classifier deployment, not as a proof of universal generalization across embedded hardware.

IV. EXPERIMENTAL SETUP

The main dataset contains 706 clean labeled events: 62 artillery launches and 644 non-launch events. We use a stratified event-level 80/20 train-test split, and we apply train-time

augmentation only after the split to prevent leakage between original events and their augmented variants. We evaluate robustness under controlled MFCC-domain perturbation at nominal SNR levels of 30, 20, 10, and 5 dB. We report accuracy, precision, recall, F1-score, and Matthews correlation coefficient (MCC), and compare:

- 1) desktop float32 post-trigger classification under controlled MFCC-space perturbation,
- 2) PC-side int8 verification of the quantized model, and
- 3) ESP32-S3 deployment of the same quantized post-trigger classifier.

V. RESULTS

A. Post-Trigger Classification Under Nominal MFCC-Space SNR

Table II summarizes the desktop float32 reference. The model retains high recall across all tested SNR levels, but precision is low and degrades further under heavier corruption. As a result, MCC remains moderate and declines from 0.5377 at 30 dB to 0.4581 at 5 dB. The float32 reference therefore fails mainly through false-positive inflation rather than missed artillery launches.

TABLE II
DESKTOP FLOAT32 RESULTS UNDER CONTROLLED MFCC-DOMAIN PERTURBATION. SNR DENOTES NOMINAL FEATURE-SPACE SNR, NOT WAVEFORM-DOMAIN ACOUSTIC SNR.

SNR (dB)	Accuracy (%)	Precision	Recall	F1-score	MCC
30	84.41	0.35	1.00	0.52	0.5377
20	82.86	0.33	1.00	0.49	0.5367
10	81.12	0.31	0.99	0.47	0.4893
5	80.44	0.29	0.93	0.45	0.4581

B. PC-Side int8 Verification

To separate hardware effects from quantization effects, we evaluate the quantized model on PC before moving to the embedded target. The PC-side int8 results in Table III are consistently stronger than the float32 reference, with MCC values of 0.9211 at 30 dB and 0.6511 at 5 dB. This intermediate step suggests that quantization contributes to the robustness improvement observed later on the embedded target, although we do not isolate the exact mechanism from all preprocessing and deployment-related factors.

C. ESP32-S3 Deployment Results

Table III also reports the embedded ESP32-S3 results for the same quantized classifier. The embedded run preserves high classification performance across all tested SNR levels and does not show harmful drift relative to the PC-side int8 reference. In this setup, the embedded target attains MCC values from 0.9912 at 30 dB to 0.8009 at 5 dB. The average per-segment inference latency is approximately 32 ms, and the deployed implementation uses an 80 KB `tensor_arena`. These results indicate that the quantized post-trigger classifier remains deployable on the target platform under the tested conditions.

TABLE III
QUANTIZED INFERENCE COMPARISON: PC-SIDE INT8 VERIFICATION VS. ESP32-S3 DEPLOYMENT. SNR DENOTES NOMINAL FEATURE-SPACE SNR, NOT WAVEFORM-DOMAIN ACOUSTIC SNR.

Mode	SNR (dB)	Accuracy (%)	Precision	Recall	F1-score	MCC
PC int8	30	98.66	0.86	1.00	0.93	0.9211
PC int8	20	96.77	0.72	1.00	0.84	0.8340
PC int8	10	92.74	0.54	0.97	0.69	0.6890
PC int8	5	92.41	0.53	0.90	0.66	0.6511
ESP32-S3	30	99.86	1.00	0.98	0.99	0.9912
ESP32-S3	20	99.73	0.98	0.98	0.98	0.9824
ESP32-S3	10	97.78	0.86	0.87	0.87	0.8554
ESP32-S3	5	97.18	0.95	0.70	0.81	0.8009

VI. DISCUSSION AND LIMITATIONS

The results support a two-stage interpretation: data acquisition and trigger generation act as upstream components, while the post-trigger classifier makes the semantic decision. The float32 desktop reference is not a sufficient proxy for embedded performance under controlled MFCC-domain perturbation. Quantized inference, first verified on PC and then deployed on ESP32-S3, stays substantially more stable under the tested nominal SNR levels.

The results also have clear limits. First, we do not evaluate continuous-stream trigger performance or false-trigger rates, so our system-level claim is limited to the post-trigger classification stage. Second, the main evaluation uses a single stratified hold-out split. We perform the split before augmentation to prevent leakage, but broader uncertainty estimation is left for future work. Third, the robustness study is based on controlled perturbation in MFCC space, and the reported nominal SNR should not be interpreted as waveform-domain acoustic SNR. Although this setup is appropriate for assessing feature-space stability of the classifier, it does not replace waveform-level acoustic SNR validation, where noise is added before MFCC extraction. Such waveform-level validation remains necessary future work. Fourth, the apparent robustness benefit associated with quantization is an empirical observation supported by the PC-side int8 bridge experiment, not an isolated mechanistic proof. Finally, the corpus remains relatively small in terms of positive artillery-launch events, and the deployment study covers ESP32-S3 rather than all embedded targets.

VII. CONCLUSION

We presented a lightweight two-stage acoustic pipeline for embedded artillery launch detection and evaluated its post-trigger classification stage under controlled MFCC-domain perturbation. The classifier uses a compact LeNet-like MFCC-CNN architecture designed to provide a small-footprint second-stage model that is simple enough for MCU deployment while still retaining useful discrimination under controlled feature-space degradation.

On the expanded 706-event corpus, the desktop float32 reference remains vulnerable to false positives, whereas quantized inference is markedly more robust under the tested nominal SNR levels. PC-side int8 verification and ESP32-S3 deployment show that the post-trigger classifier can run on

the target platform without harmful deployment drift, reaching MCC values of 0.9912 at 30 dB and 0.8009 at 5 dB with approximately 32 ms average per-segment inference latency and an 80 KB `tensor_arena`.

Within the stated scope, these results support embedded post-trigger classification as a viable component of artillery-launch acoustic sensing. The contribution therefore combines an experimentally grounded embedded validation of a practical second-stage classifier with a task-specific lightweight MFCC-CNN design. Future work should extend the validation to continuous-stream false-trigger rates, waveform-domain acoustic noise, repeated-split uncertainty estimation, and broader deployment targets.

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